CLASS 10 · CBSE · NCERT MATHEMATICS · NCERT SOLUTIONS

Chapter 10: Circles

Step-by-step solutions to the textbook exercises: 17 questions

Questions are numbered as in the textbook. Try each one yourself before reading the solution.

1 Exercise 10.1

Q1 State the number of tangents that a circle has.

SOLUTION

There is exactly one tangent at each point of a circle, and a circle has infinitely many points.

Answer: infinitely many

Q2 Complete each sentence: (i) A tangent meets a circle at exactly ___ point(s). (ii) A line that cuts a circle at two points is called a ___. (iii) At most ___ tangents to a circle can be parallel to one another. (iv) The point where a tangent touches the circle is known as the ___.

SOLUTION

(iii) Two parallel tangents touch the circle at the ends of a diameter, so there can be only two.

Answer: (i) one (ii) secant (iii) two (iv) the point of contact

Q3 A tangent PQ at a point P of a circle of radius 5 cm meets a line through the centre O at Q, with $OQ = 12$ cm. The length PQ is (A) 12 cm (B) 13 cm (C) 8.5 cm (D) $\sqrt{119}$ cm.

SOLUTION

The radius is perpendicular to the tangent at the point of contact, so $\angle OPQ = 90^\circ$.

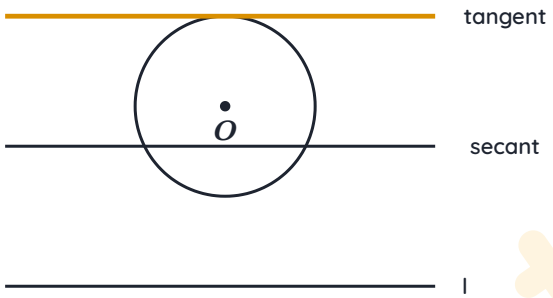
$PQ = \sqrt{OQ^2 - OP^2} = \sqrt{144 - 25} = \sqrt{119}$ cm.

Answer: (D) $\sqrt{119}$ cm

Q4 Given a line, sketch a circle and then draw two lines parallel to the given line: one that is a tangent to the circle and one that is a secant.

SOLUTION

Draw the given line l and a circle with centre O. Draw a line parallel to l that cuts the circle at two points: this is the secant. Then draw another line parallel to l that just touches the circle at one point: this is the tangent.



The tangent and the secant are both parallel to the line l

The figure shows one such tangent and secant.

2 Exercise 10.2

Q1 A point Q is 25 cm from the centre of a circle, and the tangent from Q to the circle is 24 cm long. What is the radius? (A) 7 cm (B) 12 cm (C) 15 cm (D) 24.5 cm.

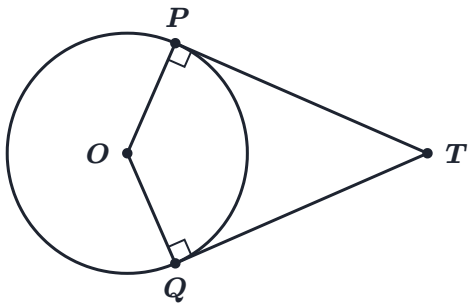
SOLUTION

The radius is perpendicular to the tangent at the point of contact, so radius $= \sqrt{25^2 - 24^2} = \sqrt{49} = 7$ cm.

Answer: (A) 7 cm

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Q2 TP and TQ are two tangents to a circle with centre O such that $\angle POQ = 110°$. Then $\angle PTQ$ is (A) $60°$ (B) $70°$ (C) $80°$ (D) $90°$.



SOLUTION

In quadrilateral $OPTQ$, $\angle OPT = \angle OQT = 90°$ (radius $\perp$ tangent). The four angles add up to $360°$.

$\angle PTQ = 360° - 90° - 90° - 110° = 70°$.

Answer: (B) $70°$

Q3 Tangents PA and PB from a point P to a circle with centre O are inclined to each other at $80°$. Then $\angle POA$ is (A) $50°$ (B) $60°$ (C) $70°$ (D) $80°$.

SOLUTION

In quadrilateral $OAPB$, $\angle OAP = \angle OBP = 90°$, so $\angle AOB = 180° - 80° = 100°$.

$\triangle OAP \cong \triangle OBP$ (RHS: $OA = OB$, OP common), so OP bisects $\angle AOB$ and $\angle POA = 50°$.

Answer: (A) $50°$

Q4 Show that the tangents at the two ends of any diameter of a circle are parallel to each other.

SOLUTION

Let AB be a diameter of a circle with centre O, and let PQ and RS be the tangents at A and B.

The radius is perpendicular to the tangent at the point of contact, so $OA \perp PQ$ and $OB \perp RS$.

That is, $\angle PAB = 90°$ and $\angle ABS = 90°$.

These are alternate interior angles made by the transversal AB with PQ and RS, and they are equal.

Hence $PQ \parallel RS$.

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Q5 Show that the line drawn perpendicular to a tangent at its point of contact must pass through the centre of the circle.

SOLUTION

Let the tangent AB touch the circle with centre O at P. We know that $OP \perp AB$, because the radius through the point of contact is perpendicular to the tangent.

At a given point of a line, only one perpendicular to the line can be drawn. So the perpendicular to AB at P is the line OP itself.

Hence the perpendicular at the point of contact passes through the centre O.

Q6 The length of a tangent from a point A, 5 cm from the centre of a circle, is 4 cm. Find the radius of the circle.

SOLUTION

Let the tangent touch the circle at B, with centre O. Then $\angle OBA = 90^\circ$, $OA = 5$ cm and $AB = 4$ cm.

$OB = \sqrt{5^2 - 4^2} = 3$ cm.

Answer: 3 cm

Q7 Two concentric circles have radii 5 cm and 3 cm. Find the length of the chord of the larger circle that touches the smaller circle.

SOLUTION

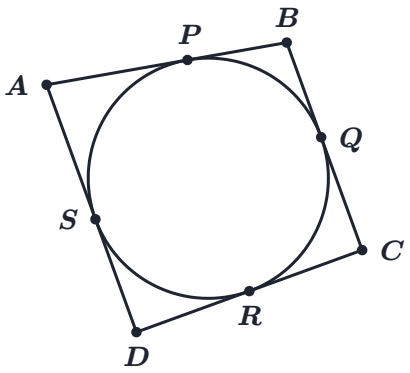
Let the chord AB of the larger circle touch the smaller circle at M, with centre O. Then $OM \perp AB$, $OM = 3$ cm and $OA = 5$ cm.

$AM = \sqrt{5^2 - 3^2} = 4$ cm. The perpendicular from the centre bisects the chord, so $AB = 2 \times 4 = 8$ cm.

Answer: 8 cm

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Q8 A quadrilateral $ABCD$ is drawn to circumscribe a circle. Prove that $AB + CD = AD + BC$.



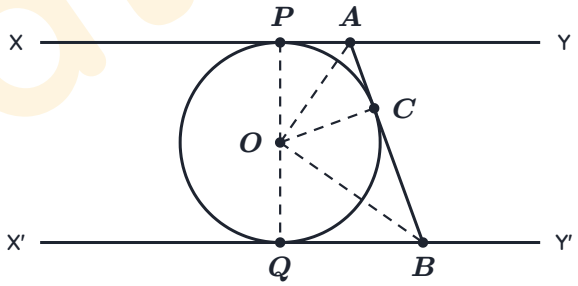
SOLUTION

Let the sides AB, BC, CD, DA touch the circle at P, Q, R, S. Tangents from an external point to a circle are equal, so $AP = AS$, $BP = BQ$, $CR = CQ$ and $DR = DS$.

Adding: $(AP + BP) + (CR + DR) = (AS + DS) + (BQ + CQ)$.

Hence $AB + CD = AD + BC$.

Q9 XY and $X'Y'$ are two parallel tangents to a circle with centre O. Another tangent AB, with point of contact C, meets XY at A and $X'Y'$ at B. Prove that $\angle AOB = 90°$.



SOLUTION

Let XY touch the circle at P and $X'Y'$ touch it at Q. Join OC.

In $\triangle OPA$ and $\triangle OCA$: $OP = OC$ (radii), $AP = AC$ (tangents from A) and OA is common. So $\triangle OPA \cong \triangle OCA$ (SSS), and $\angle POA = \angle COA$.

In the same way $\triangle OQB \cong \triangle OCB$, so $\angle QOB = \angle COB$.

The tangents at P and Q are parallel, so PQ is a diameter and POQ is a straight line: $\angle POA + \angle COA + \angle COB + \angle QOB = 180°$.

So $2(\angle COA + \angle COB) = 180°$, that is $\angle COA + \angle COB = 90°$.

Hence $\angle AOB = 90°$.

Q10 Two tangents are drawn to a circle from a point outside it. Prove that the angle between the tangents and the angle that the chord of contact makes at the centre add up to $180°$.

SOLUTION

Let PA and PB be the tangents from P to a circle with centre O. Then $\angle OAP = \angle OBP = 90°$ (radius $\perp$ tangent).

In quadrilateral $OAPB$: $\angle APB + \angle AOB + 90° + 90° = 360°$.

Hence $\angle APB + \angle AOB = 180°$, so the two angles are supplementary.

Q11 Prove that a parallelogram circumscribing a circle is a rhombus.

SOLUTION

Let $ABCD$ be a parallelogram circumscribing a circle. Then $AB = CD$ and $AD = BC$ (opposite sides of a parallelogram).

For any quadrilateral circumscribing a circle, $AB + CD = AD + BC$ (proved in Q8).

So $2AB = 2AD$, that is $AB = AD$. Then $AB = BC = CD = DA$.

A parallelogram with all sides equal is a rhombus.

Q12 A triangle ABC circumscribes a circle of radius 4 cm. The point of contact D divides BC into $BD = 8$ cm and $DC = 6$ cm. Find AB and AC.

SOLUTION

Let the circle touch AB at F and AC at E. Tangents from a point are equal: $BF = BD = 8$ cm, $CE = CD = 6$ cm, and let $AF = AE = x$ cm.

Then $AB = 8 + x$, $AC = 6 + x$, $BC = 14$, and the semi-perimeter is $s = 14 + x$.

By Heron's formula, area $= \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{(14+x) \times x \times 8 \times 6} = \sqrt{48x(14+x)}$.

Joining the centre O to the vertices splits the triangle into three triangles of height 4 cm: area $= \frac{1}{2} \times 4 \times (AB + BC + CA) = 4(14 + x)$.

So $48x(14+x) = 16(14+x)^2$, which gives $3x = 14 + x$ and $x = 7$.

Answer: $AB = 15$ cm and $AC = 13$ cm

Q13 A quadrilateral is drawn around a circle so that all four of its sides touch the circle. Prove that any two opposite sides make angles at the centre that add up to 180°.

SOLUTION

Let $ABCD$ touch the circle with centre O at P, Q, R, S on AB, BC, CD, DA (as in the figure of Q8). Join O to the four vertices and the four points of contact.

$\triangle AOP \cong \triangle AOS$ (SSS: $OP = OS$, $AP = AS$, OA common), so $\angle AOP = \angle AOS$. In the same way $\angle BOP = \angle BOQ$, $\angle COQ = \angle COR$ and $\angle DOR = \angle DOS$.

The eight angles at O add up to 360°, so $2(\angle AOP + \angle BOP + \angle COR + \angle DOR) = 360^\circ$.

That is $(\angle AOP + \angle BOP) + (\angle COR + \angle DOR) = 180^\circ$, so $\angle AOB + \angle COD = 180^\circ$. Similarly $\angle BOC + \angle AOD = 180^\circ$.

Hence opposite sides subtend supplementary angles at the centre.

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